

COMBINATION AND CUTTING OF SPRINGS

JEE Advanced Preparation — Physics

Topic 01 | Oscillations & Elasticity | Spring Constant, Series, Parallel, Cutting, SHM

Concept	Formula	Key Point
Hooke's Law	$F = kx$	k = spring constant (N/m)
Series	$1/k_{\text{eff}} = 1/k_1 + 1/k_2$	Same force, extensions add
Parallel	$k_{\text{eff}} = k_1 + k_2$	Same extension, forces add
Cut (n equal)	$k_{\text{new}} = n \cdot k$	k inversely prop. to length
Cut (ratio m:n)	$k_1 = k(m+n)/m$	$k \cdot L$ = constant rule
SHM period	$T = 2\pi \sqrt{m/k}$	Independent of amplitude

This module provides a comprehensive study of spring systems for JEE Advanced: from Hooke's law derivation to complex mixed networks. All standard question types are covered with full worked solutions and 10 JEE-style MCQs.

KEY FORMULAS REFERENCE

- Hooke's Law:** $F = k \cdot x$ (restoring force, N; k in N/m; x = deformation in m)
- Series combination:** $1/k_{\text{eff}} = 1/k_1 + 1/k_2 + 1/k_3 + \dots$ (same force, extensions add)
- Parallel combination:** $k_{\text{eff}} = k_1 + k_2 + k_3 + \dots$ (same extension, forces add)
- Cutting rule:** $k \cdot L = \text{constant}$ (spring constant $\times$ natural length = constant)
- Cut into n equal parts:** each piece has $k_{\text{piece}} = n \cdot k$
- Cut in ratio m:n $\Rightarrow$** $k_1 = k(m+n)/m$, $k_2 = k(m+n)/n$
- SHM:** $\omega = \sqrt{k_{\text{eff}}/m}$, $T = 2\pi \sqrt{m/k_{\text{eff}}}$, $f = (1/2\pi) \sqrt{k_{\text{eff}}/m}$
- Max velocity:** $v_{\text{max}} = A \cdot \omega = A \cdot \sqrt{k/m}$ (at mean/equilibrium position)
- Max acceleration:** $a_{\text{max}} = A \cdot \omega^2 = A \cdot k/m$ (at extreme positions)
- Total energy:** $E = (1/2) \cdot k \cdot A^2$ (constant; independent of position)

SECTION 1: HOOKE'S LAW AND SPRING CONSTANT

A spring is an elastic device that stores mechanical energy when deformed (stretched or compressed). The fundamental law governing spring behavior is Hooke's Law, valid within the elastic limit of the material.

Hooke's Law: $F = -k \cdot x$

where: F = restoring force exerted by spring (N), k = spring constant (N/m), x = displacement from natural length (m).

The negative sign indicates the force is always directed towards the equilibrium position (restoring nature).

Physical interpretation of k :

k measures the stiffness of the spring. A spring with $k = 1000$ N/m needs 1000 N to stretch it by 1 metre.

Stiffer spring $\Rightarrow$ larger k . More flexible spring $\Rightarrow$ smaller k .

Factors affecting k for a helical spring:

$k = \frac{G \cdot d^4}{8 \cdot D^3 \cdot n}$ where G = shear modulus, d = wire diameter, D = mean coil diameter, n = number of coils.

From this formula: k is inversely proportional to n (number of coils) and to D^3 , proportional to d^4 .

Since n is proportional to the natural length L : k is inversely proportional to L .

Therefore: $k \cdot L = \text{constant}$ (the fundamental cutting rule).

Units and dimensions:

k : N/m = kg/s² [$MLT^{-2} / L = MT^{-2}$]

Dimension of k : [MT^{-2}]

NOTE: Hooke's Law is valid only within the proportionality limit. Beyond the elastic limit, the spring permanently deforms and k is no longer constant. JEE problems always assume ideal springs.

SECTION 2: SERIES COMBINATION OF SPRINGS

Springs are in series when they are connected end-to-end such that the force transmitted through all springs is the same.

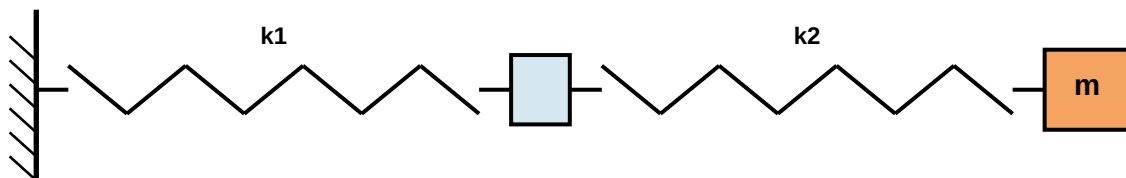


Fig 1: Springs k_1 and k_2 in Series — same tension, extensions add

Setup: Springs with constants $k_1, k_2, \dots$ are connected end-to-end. A force F is applied.

Key property: Same force acts on each spring (tension = F throughout).

Proof for two springs in series:

Let x_1 = extension of spring 1, x_2 = extension of spring 2.

Since force is same: $F = k_1 x_1 \Rightarrow x_1 = F/k_1$

$F = k_2 x_2 \Rightarrow x_2 = F/k_2$

Total extension: $x_{\text{total}} = x_1 + x_2 = F/k_1 + F/k_2 = F(1/k_1 + 1/k_2)$

But $x_{\text{total}} = F/k_{\text{eff}}$, so: $1/k_{\text{eff}} = 1/k_1 + 1/k_2$

For n springs in series:

$1/k_{\text{eff}} = 1/k_1 + 1/k_2 + \dots + 1/k_n$

Special cases:

If all n springs are identical (each = k): $1/k_{\text{eff}} = n/k \Rightarrow k_{\text{eff}} = k/n$

For two equal springs: $k_{\text{eff}} = k/2$

k_{eff} is ALWAYS less than the smallest k in the series.

Distribution of extensions in series:

$x_1/x_2 = k_2/k_1$ (spring with smaller k stretches more)

Longer spring (smaller k) takes larger share of total extension.

SECTION 3: PARALLEL COMBINATION OF SPRINGS

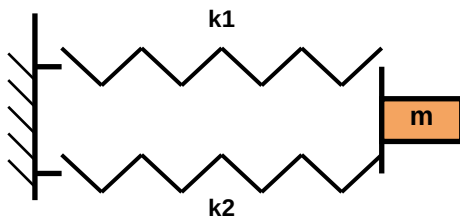


Fig 2: Springs in Parallel — same extension, forces add

Setup: Springs are connected side-by-side, both attached to the same two points.

Key property: Same extension (displacement) in each spring.

Proof for two springs in parallel:

Both springs extend by the same amount x (constraint).

Force in spring 1: $F_1 = k_1 x$

Force in spring 2: $F_2 = k_2 x$

Total force: $F = F_1 + F_2 = (k_1 + k_2)x$

Since $F = k_{\text{eff}}x$: $k_{\text{eff}} = k_1 + k_2$

For n springs in parallel:

$k_{\text{eff}} = k_1 + k_2 + \dots + k_n$

Special cases:

n identical springs: $k_{\text{eff}} = n k$

k_{eff} is ALWAYS greater than the largest k in the combination.

Distribution of forces in parallel:

$F_1/F_2 = k_1/k_2$ (stiffer spring bears larger share of load)

Comparison: Series vs Parallel

Series: $k_{\text{eff}} < k_{\text{min}}$ (softer combined system)

Parallel: $k_{\text{eff}} > k_{\text{max}}$ (stiffer combined system)

A given mass oscillates faster on parallel springs, slower on series springs.

SECTION 4: CUTTING OF SPRINGS — DETAILED ANALYSIS

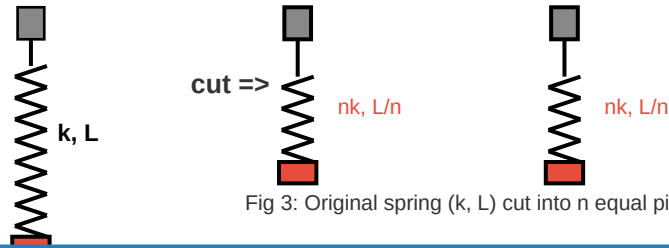


Fig 3: Original spring (k, L) cut into n equal pieces (each: nk, L/n)

Fundamental Rule: $k \cdot L = \text{constant}$ (for any segment of the same spring)

This follows because each coil contributes equally, and cutting changes the number of coils.

Case 1: Cut into n equal parts

Original spring: natural length L, constant k.

Each piece: length L/n .

New constant: $k_{\text{new}} = k \cdot L / (L/n) = n \cdot k$

Example: spring $k=100$ N/m cut into 5 equal pieces $\Rightarrow$ each piece has $k_{\text{new}} = 500$ N/m

Case 2: Cut in ratio m:n (m+n parts total)

Piece 1: length $L_1 = m \cdot L / (m+n)$, constant $k_1 = k \cdot L / L_1 = k \cdot (m+n) / m$

Piece 2: length $L_2 = n \cdot L / (m+n)$, constant $k_2 = k \cdot L / L_2 = k \cdot (m+n) / n$

Verification: $1/k_1 + 1/k_2 = m / (k \cdot (m+n)) + n / (k \cdot (m+n)) = (m+n) / (k \cdot (m+n)) = 1/k$ [series check OK]

Case 3: General — cut at length fraction f ($0 < f < 1$)

Piece 1: length $f \cdot L$, constant k/f

Piece 2: length $(1-f) \cdot L$, constant $k/(1-f)$

Important applications:

1. Cut a spring and reconnect in series $\Rightarrow$ same as original ($1/k_1 + 1/k_2 = 1/k$)
2. Cut and reconnect in parallel $\Rightarrow k_{\text{eff}} = k_1 + k_2 = k \cdot (m+n) \cdot (1/m + 1/n) = k \cdot (m+n)^2 / (m \cdot n) > k$
3. Using all n equal pieces in parallel: $k_{\text{eff_parallel}} = n \cdot k_{\text{new}} = n \cdot (n \cdot k) = n^2 \cdot k$

IMPORTANT: The rule ' k proportional to $1/L$ ' applies ONLY to pieces of the SAME spring.

Different springs have different constants regardless of length.

For a spring of constant k and length L : energy stored $= (1/2) \cdot k \cdot x^2 = (1/2) \cdot (kL) \cdot (x^2/L) = \text{constant} \cdot \text{strain}^2$

SECTION 5: TIME PERIOD OF SPRING-MASS OSCILLATION

A mass-spring system executes Simple Harmonic Motion (SHM) when displaced from equilibrium. Understanding the time period is essential for JEE Advanced oscillation problems.

Derivation of SHM equation:

Let x = displacement from equilibrium position (natural length + static extension).

Net restoring force: $F = -k_{\text{eff}} * x$ (Hooke's law, directed toward equilibrium)

By Newton's second law: $m * (d^2x/dt^2) = -k_{\text{eff}} * x$

$\Rightarrow d^2x/dt^2 = -(k_{\text{eff}}/m) * x$

This is SHM with: $\omega^2 = k_{\text{eff}}/m$

$\Rightarrow \omega = \sqrt{k_{\text{eff}}/m}$ (angular frequency in rad/s)

Time period:

$T = 2\pi / \omega = 2\pi * \sqrt{m/k_{\text{eff}}}$

Frequency:

$f = 1/T = (1/(2\pi)) * \sqrt{k_{\text{eff}}/m}$ (oscillations per second, Hz)

Important: Effect of gravity

For a vertical spring-mass system, gravity only shifts the equilibrium position.

It does NOT change the time period. $T = 2\pi\sqrt{m/k_{\text{eff}}}$ regardless of orientation.

Proof: Let $x_0 = mg/k$ (static extension). New displacement $y = x - x_0$.

Equation: $m\ddot{y} = -k\ddot{y}$ (gravity terms cancel). Same SHM as without gravity.

Special cases for T:

Series: $T_s = 2\pi\sqrt{m/k_s} = 2\pi\sqrt{m*(1/k_1+1/k_2)} > T_{\text{individual}}$

Parallel: $T_p = 2\pi\sqrt{m/(k_1+k_2)} < T_{\text{individual}}$

$T_{\text{series}} / T_{\text{parallel}} = \sqrt{k_p/k_s} = \sqrt{(k_1+k_2)*(1/k_1+1/k_2)} \geq 2$

SECTION 6: ENERGY IN SPRING-MASS SYSTEM

Elastic Potential Energy: $PE = (1/2)kx^2$

Kinetic Energy: $KE = (1/2)mv^2$

Total Energy (conserved): $E = (1/2)kA^2$ (A = amplitude)

At any position x:

$$KE = E - PE = (1/2)k(A^2 - x^2)$$

$$v = \sqrt{k/m} \sqrt{A^2 - x^2} = \omega \sqrt{A^2 - x^2}$$

At mean position (x=0): $KE = E = (1/2)kA^2$, $PE = 0$, $v = A\omega$ (maximum)

At extreme position (x=A): $KE = 0$, $PE = E = (1/2)kA^2$, $v = 0$

At $x = A/\sqrt{2}$: $KE = PE = E/2$ (energy equally divided)

Power: $P = Fv = -kx \cdot v = -kx \cdot \omega \sqrt{A^2 - x^2}$

Average power over a complete cycle = 0 (conservative force)

Spring balance in various situations:

Spring balance reads tension in the string/spring, NOT weight of the body.

If body is accelerating upward at a: reading = $m(g+a)$

If body is in free fall: reading = 0 (weightlessness)

If body is on smooth incline (angle θ): reading = $m g \sin(\theta)$

SECTION 7: MIXED SPRING NETWORKS

Strategy for complex networks:

Step 1: Identify nodes (junction points) in the spring network.

Step 2: Springs sharing the SAME two nodes are in PARALLEL.

Step 3: Springs with no intermediate mass, connected end-to-end, are in SERIES.

Step 4: Reduce iteratively: replace each parallel group and series group with k_{eff} .

Step 5: Repeat until a single equivalent spring remains.

Common configurations and their k_{eff} :

3 springs k_1, k_2, k_3 all in series: $1/k_{\text{eff}} = 1/k_1 + 1/k_2 + 1/k_3$

3 equal springs k in series: $k_{\text{eff}} = k/3$

k_1 and k_2 in parallel, in series with k_3 : $k_{\text{eff}} = (k_1+k_2)*k_3/(k_1+k_2+k_3)$

k_1 in series with (k_2 parallel k_3): $k_{\text{eff}} = k_1*(k_2+k_3)/(k_1+k_2+k_3)$

Network reduction example:

Given: $k_1=2, k_2=3, k_3=6$ (N/m). k_1 and k_2 in parallel, then in series with k_3 .

Step 1: $k_{12} = k_1+k_2 = 5$ N/m

Step 2: $1/k_{\text{eff}} = 1/k_{12} + 1/k_3 = 1/5 + 1/6 = 11/30$

Step 3: $k_{\text{eff}} = 30/11$ approximately 2.73 N/m

Wheatstone bridge analogy for springs:

For a balanced bridge ($k_1/k_2 = k_3/k_4$), the middle spring carries no force.

This simplifies analysis: remove middle spring, solve remaining series/parallel combination.

WORKED EXAMPLES — JEE ADVANCED LEVEL

Example 1: Two Springs in Series

Problem: Springs $k_1=2k$ and $k_2=3k$ are connected in series. Find k_{eff} .

Solution:

$$1/k_{\text{eff}} = 1/k_1 + 1/k_2 = 1/(2k) + 1/(3k)$$

$$1/k_{\text{eff}} = 3/(6k) + 2/(6k) = 5/(6k)$$

$$k_{\text{eff}} = 6k/5$$

Check: $k_{\text{eff}} = 6k/5 = 1.2k$, which is less than $k_1=2k$ and $k_2=3k$. Correct for series.

If mass m is attached, $T = 2\pi\sqrt{m/k_{\text{eff}}} = 2\pi\sqrt{5m/(6k)}$

Example 2: Spring Cut and Parallel Combination

Problem: Spring $k=100$ N/m cut into 4 equal parts. Three parts joined in parallel. Find k_{eff} .

Solution:

Step 1: Each of 4 equal parts: $k_{\text{part}} = 4 * 100 = 400$ N/m [cutting rule: $k_{\text{new}} = n*k$]

Step 2: Three parts in parallel: $k_{\text{eff}} = 400 + 400 + 400 = 1200$ N/m

Step 3: If mass $m = 0.3$ kg: $T = 2\pi\sqrt{0.3/1200} = 2\pi\sqrt{2.5e-4} = 2\pi*0.0158 = 0.0993$ s

Answer: $k_{\text{eff}} = 1200$ N/m

Example 3: Time Period Ratio — Series vs Parallel

Problem: Springs k and $2k$; mass m . Compare T in series vs T in parallel.

Solution:

Series: $1/k_s = 1/k + 1/(2k) = 3/(2k) \Rightarrow k_s = 2k/3$

Parallel: $k_p = k + 2k = 3k$

$T_{\text{series}} = 2\pi\sqrt{m/k_s} = 2\pi\sqrt{3m/(2k)}$

$T_{\text{parallel}} = 2\pi\sqrt{m/k_p} = 2\pi\sqrt{m/(3k)}$

$T_{\text{series}}/T_{\text{parallel}} = \sqrt{k_p/k_s} = \sqrt{3k / (2k/3)} = \sqrt{9/2} = 3/\sqrt{2} = 3\sqrt{2}/2$
approx 2.12

So the series arrangement oscillates more than twice as slowly as the parallel arrangement.

Example 4: Spring Cut in Ratio 1:2, Then Parallel

Problem: Spring k cut in ratio 1:2. Two pieces joined in parallel. Find k_{eff} .

Solution:

$m=1, n=2$: Total parts = 3.

$k_1 = k*(1+2)/1 = 3k$ (shorter piece, length $L/3$)

$k_2 = k*(1+2)/2 = 3k/2$ (longer piece, length $2L/3$)

Verification: $1/k_1 + 1/k_2 = 1/(3k) + 2/(3k) = 3/(3k) = 1/k$ [checks out]

Parallel: $k_{\text{eff}} = k_1 + k_2 = 3k + 3k/2 = 6k/2 + 3k/2 = 9k/2$

Answer: $k_{\text{eff}} = 9k/2 = 4.5k$

Example 5: Mixed Network — Three Springs

Problem: $k_1=k, k_2=k$ in parallel, result in series with $k_3=k$. Find k_{eff} .

Solution:

Step 1: $k_{12} = k_1 + k_2 = k + k = 2k$ (parallel combination)

Step 2: k_{12} and k_3 in series: $1/k_{\text{eff}} = 1/(2k) + 1/k = 1/(2k) + 2/(2k) = 3/(2k)$

$k_{\text{eff}} = 2k/3$

Step 3: $T = 2\pi\sqrt{m/k_{\text{eff}}} = 2\pi\sqrt{3m/(2k)}$

Answer: $k_{\text{eff}} = 2k/3$. System is softer than any individual spring ($k_{\text{eff}} < k$).

Example 6: Complete SHM Analysis

Problem: Mass $m=0.25$ kg, spring $k=400$ N/m, amplitude $A=10$ cm on smooth surface.

Find: (a) ω , (b) T , (c) f , (d) v_{max} , (e) a_{max} , (f) total energy.

Solution:

(a) $\omega = \sqrt{k/m} = \sqrt{400/0.25} = \sqrt{1600} = 40$ rad/s

(b) $T = 2\pi/\omega = 2*3.14159/40 = 0.157$ s

(c) $f = 1/T = 1/0.157 = 6.37$ Hz

(d) $v_{\text{max}} = A*\omega = 0.10*40 = 4.0$ m/s (at equilibrium)

(e) $a_{\text{max}} = A*\omega^2 = 0.10*1600 = 160$ m/s² (at extremes)

(f) $E = (1/2)*k*A^2 = 0.5*400*(0.10)^2 = 0.5*400*0.01 = 2.0$ J

PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. A spring of spring constant k is cut into three equal parts and these parts are connected in parallel. The effective spring constant of the combination is:

(A) $k/3$ (B) k (C) $3k$ (D) $9k$

Q2. Two springs A ($k_A=2k$) and B ($k_B=k$) are connected in series. A mass m hangs from the combination. The ratio of extension in A to extension in B is:

(A) 2:1 (B) 1:2 (C) 1:1 (D) 3:1

Q3. A spring of natural length L and constant k is cut such that one piece has length $L/4$. The spring constant of this shorter piece is:

(A) $k/4$ (B) k (C) $2k$ (D) $4k$

Q4. A mass m is connected to two springs k_1 and k_2 in parallel on a smooth surface. The angular frequency of oscillation is:

(A) $\sqrt{(k_1+k_2)/m}$ (B) $\sqrt{m/(k_1+k_2)}$ (C) $\sqrt{k_1 k_2 / (m(k_1+k_2))}$ (D) $\sqrt{(k_1+k_2)*m}$

Q5. In a series spring combination (constants k_1 and k_2), the ratio of extensions x_1/x_2 is:

(A) k_1/k_2 (B) k_2/k_1 (C) $(k_1+k_2)/k_1$ (D) 1

Q6. A spring is compressed by $x = 10$ cm and released. The attached block ($m=0.5$ kg, $k=200$ N/m) achieves maximum speed:

(A) 1 m/s (B) 2 m/s (C) $\sqrt{2}$ m/s (D) 4 m/s

Q7. Three springs of constant k each: two in series, this in parallel with a third spring. The effective constant is:

(A) $5k/2$ (B) $3k/2$ (C) $2k/3$ (D) k

Q8. A spring is stretched 5 cm by a load. If it is cut into 5 equal parts and one part is loaded with the same force, the extension is:

(A) 1 cm (B) 5 cm (C) 25 cm (D) 0.2 cm

Q9. Two identical springs each of constant k are connected first in series and then in parallel with the same mass m . The ratio $T_{\text{series}}/T_{\text{parallel}}$ is:

(A) 1 (B) 2 (C) $\sqrt{2}$ (D) $1/2$

Q10. A spring balance reads 50 N when a block is suspended from it. If the same balance is used with the spring below the block (inverted), what does it read?

(A) 0 N (B) 50 N (C) 25 N (D) 100 N

ANSWER KEY AND EXPLANATIONS

Q1 — Answer: (D) 9k

Cut into 3 equal parts: each has $k_{\text{piece}} = 3k$ (using $k_{\text{new}} = n \cdot k$, $n=3$).

Three pieces in parallel: $k_{\text{eff}} = 3k + 3k + 3k = 9k$.

Check: cutting n parts then all in parallel always gives $n^2 \cdot k$. Here: $3^2 \cdot k = 9k$.

Q2 — Answer: (B) 1:2

In series, tension is same F in both springs. $x = F/k$.

$x_A = F/(2k)$, $x_B = F/k$.

$x_A/x_B = (F/2k)/(F/k) = 1/2$. So ratio is 1:2.

The softer spring (smaller k) extends more.

Q3 — Answer: (D) 4k

Piece length = $L/4$. Using $k \cdot L = k' \cdot (L/4)$: $k' = 4k$.

Alternatively: shorter spring = stiffer. k is inversely proportional to length.

Q4 — Answer: (A) $\sqrt{(k_1+k_2)/m}$

Parallel: $k_{\text{eff}} = k_1+k_2$. $\omega = \sqrt{k_{\text{eff}}/m} = \sqrt{(k_1+k_2)/m}$.

Q5 — Answer: (B) k_2/k_1

In series: $F = k_1 \cdot x_1 = k_2 \cdot x_2$. So $x_1 = F/k_1$, $x_2 = F/k_2$.

$x_1/x_2 = (F/k_1)/(F/k_2) = k_2/k_1$.

Q6 — Answer: (B) 2 m/s

$v_{\text{max}} = A \cdot \omega = A \cdot \sqrt{k/m} = 0.10 \cdot \sqrt{200/0.5} = 0.10 \cdot \sqrt{400} = 0.10 \cdot 20 = 2 \text{ m/s}$.

Q7 — Answer: (B) $3k/2$

Two springs k in series: $k_{\text{series}} = k/2$.

$k/2$ in parallel with k : $k_{\text{eff}} = k/2 + k = 3k/2$.

Q8 — Answer: (A) 1 cm

One of 5 equal parts has $k_{\text{new}} = 5k$. Under same force F :

$x' = F/k_{\text{new}} = F/(5k) = (1/5) \cdot (F/k) = (1/5) \cdot 5 \text{ cm} = 1 \text{ cm}$.

Stiffer spring extends less.

Q9 — Answer: (B) 2

Series: $k_s = k/2$. Parallel: $k_p = 2k$.

$T = 2 \cdot \pi \cdot \sqrt{m/k}$. $T_s/T_p = \sqrt{k_p/k_s} = \sqrt{2k/(k/2)} = \sqrt{4} = 2$.

Q10 — Answer: (B) 50 N

A spring balance reads the tension in the spring, which equals the weight of the hanging block.

Tension = weight = 50 N regardless of orientation (as long as static equilibrium).

Inverting the balance doesn't change the fact that the spring supports 50 N.

End of Module 01: Combination and Cutting of Springs | JEE Advanced Physics